Deep Interest Network for Click-Through Rate Prediction

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ABSTRACT

Click-through rate prediction is an essential task in industrial applications, such as online advertising. Recently deep learning based models have been proposed, which follow a similar Embedding&MLP paradigm. In these methods large scale sparse input features are first mapped into low dimensional embedding vectors, and then transformed into fixed-length vectors in a group-wise manner, finally concatenated together to fed into a multilayer perceptron (MLP) to learn the nonlinear relations among features. In this way, user features are compressed into a fixed-length representation vector, in regardless of what candidate ads are. The use of fixed-length vector will be a bottleneck, which brings difficulty for Embedding&MLP methods to capture user's diverse interests effectively from rich historical behaviors. In this paper, we propose a novel model: Deep Interest Network (DIN) which tackles this challenge by designing a local activation unit to adaptively learn the representation of user interests from historical behaviors with respect to a certain ad. This representation vector varies over different ads, improving the expressive ability of model greatly. Besides, we develop two techniques: mini-batch aware regularization and data adaptive activation function which can help training industrial deep networks with hundreds of millions of parameters. Experiments on two public datasets as well as an Alibaba real production dataset with over 2 billion samples demonstrate the effectiveness of proposed approaches, which achieve superior performance compared with state-of-the-art methods. DIN now has been successfully deployed in the online display advertising system in Alibaba, serving the main traffic.

CCS CONCEPTS

• **Information systems → Display advertising**; **Recommender systems**;

KEYWORDS

Click-Through Rate Prediction, Display Advertising, E-commerce

1 INTRODUCTION

In cost-per-click (CPC) advertising system, advertisements are ranked by the eCPM (effective cost per mille), which is the product of the bid price and CTR (click-through rate), and CTR needs to be predicted by the system. Hence, the performance of CTR prediction model has a direct impact on the final revenue and plays a key role in the advertising system. Modeling CTR prediction has received much attention from both research and industry community.

Recently, inspired by the success of deep learning in computer vision [14] and natural language processing [1], deep learning based methods have been proposed for CTR prediction task [3, 4, 21, 26]. These methods follow a similar Embedding&MLP paradigm: large scale sparse input features are first mapped into low dimensional embedding vectors, and then transformed into fixed-length vectors in a group-wise manner, finally concatenated together to fed into fully connected layers (also known as multilayer perceptron, MLP) to learn the nonlinear relations among features. Compared with commonly used logistic regression model [19], these deep learning methods can reduce a lot of feature engineering jobs and enhance the model capability greatly. For simplicity, we name these methods Embedding&MLP in this paper, which now have become popular on CTR prediction task.

However, the user representation vector with a limited dimension in Embedding&MLP methods will be a bottleneck to express user's diverse interests. Take display advertising in e-commerce site as an example. Users might be interested in different kinds of goods simultaneously when visiting the e-commerce site. That is to say, user interests are ***diverse***. When it comes to CTR prediction task, user interests are usually captured from user behavior data. Embedding&MLP methods learn the representation of all interests for a certain user by transforming the embedding vectors of user behaviors into a fixed-length vector, which is in an euclidean space where all users' representation vectors are. In other words, diverse interests of the user are compressed into a fixed-length vector, which limits the expressive ability of Embedding&MLP methods. To make the representation capable enough for expressing user's diverse interests, the dimension of the fixed-length vector needs to be largely expanded. Unfortunately, it will dramatically enlarge the size of learning parameters and aggravate the risk of overfitting under limited data. Besides, it adds the burden of computation and storage, which may not be tolerated for an industrial online system.

On the other hand, it is not necessary to compress all the diverse interests of a certain user into the same vector when predicting a candidate ad because only part of user's interests will influence his/her action (to click or not to click). For example, a female swimmer will click a recommended goggle mostly due to the bought of bathing suit rather than the shoes in her last week's shopping list. Motivated by this, we propose a novel model: Deep Interest Network (DIN), which adaptively calculates the representation vector of user interests by taking into consideration the relevance of historical behaviors given a candidate ad. By introducing a local activation unit, DIN pays attentions to the related user interests by soft-searching for relevant parts of historical behaviors and takes a weighted sum pooling to obtain the representation of user interests with respect to the candidate ad. Behaviors with higher relevance to the candidate ad get higher activated weights and dominate the representation of user interests. We visualize this phenomenon in the experiment section. In this way, the representation vector of

user interests varies over different ads, which improves the expressive ability of model under limited dimension and enables DIN to better capture user's diverse interests.

Training industrial deep networks with large scale sparse features is of great challenge. For example, SGD based optimization methods only update those parameters of sparse features appearing in each mini-batch. However, adding with traditional ℓ_2 regularization, the computation turns to be unacceptable, which needs to calculate L2-norm over the whole parameters (with size scaling up to billions in our situation) for each mini-batch. In this paper, we develop a novel mini-batch aware regularization where only parameters of non-zero features appearing in each mini-batch participate in the calculation of L2-norm, making the computation acceptable. Besides, we design a data adaptive activation function, which generalizes commonly used PReLU[12] by adaptively adjusting the rectified point w.r.t. distribution of inputs and is shown to be helpful for training industrial networks with sparse features.

The contributions of this paper are summarized as follows:

- We point out the limit of using fixed-length vector to express user's diverse interests and design a novel deep interest network (DIN) which introduces a local activation unit to adaptively learn the representation of user interests from historical behaviors w.r.t. given ads. DIN can improve the expressive ability of model greatly and better capture the diversity characteristic of user interests.
- We develop two novel techniques to help training industrial deep networks: i) a mini-batch aware regularizer, which saves heavy computation of regularization on deep networks with huge number of parameters and is helpful for avoiding overfitting, ii) a data adaptive activation function, which generalizes PReLU by considering the distribution of inputs and shows well performance.
- We conduct extensive experiments on both public and Alibaba datasets. Results verify the effectiveness of proposed DIN and training techniques. Our code[1] is publicly available. The proposed approaches have been deployed in the commercial display advertising system in Alibaba, one of world's largest advertising platform, contributing significant improvement to the business.

In this paper we focus on the CTR prediction modeling in the scenario of display advertising in e-commerce industry. Methods discussed here can be applied in similar scenarios with rich user behaviors, such as personalized recommendation in e-commerce sites, feeds ranking in social networks etc.

The rest of the paper is organized as follows. We discuss related work in section 2 and introduce the background about characteristic of user behavior data in display advertising system of e-commerce site in section 3. Section 4 and 5 describe in detail the design of DIN model as well as two proposed training techniques. We present experiments in section 6 and conclude in section 7.

2 RELATEDWORK

The structure of CTR prediction model has evolved from shallow to deep. At the same time, the number of samples and the dimension of features used in CTR model have become larger and larger. In order to better extract feature relations to improve performance, several works pay attention to the design of model structure.

As a pioneer work, NNLM [2] learns distributed representation for each word, aiming to avoid curse of dimension in language modeling. This method, often referred to as embedding, has inspired many natural language models and CTR prediction models that need to handle large-scale sparse inputs.

LS-PLM [9] and FM [20] models can be viewed as a class of networks with one hidden layer, which first employs embedding layer on sparse inputs and then imposes specially designed transformation functions for target fitting, aiming to capture the combination relations among features.

Deep Crossing [21], Wide&Deep Learning [4] and YouTube Recommendation CTR model [3] extend LS-PLM and FM by replacing the transformation function with complex MLP network, which enhances the model capability greatly. PNN[5] tries to capture high-order feature interactions by involving a product layer after embedding layer. DeepFM[10] imposes a factorization machines as "wide" module in Wide&Deep [4] with no need of feature engineering. Overall, these methods follow a similar model structure with combination of embedding layer (for learning the dense representation of sparse features) and MLP (for learning the combination relations of features automatically). This kind of CTR prediction model reduces the manual feature engineering jobs greatly. Our base model follows this kind of model structure. However in applications with rich user behaviors, features are often contained with variable-length list of ids, e.g., searched terms or watched videos in YouTube recommender system [3]. These models often transform corresponding list of embedding vectors into a fixed-length vector via sum/average pooling, which causes loss of information. The proposed DIN tackles it by adaptively learning the representation vector w.r.t. given ad, improving the expressive ability of model.

Attention mechanism originates from Neural Machine Translation (NMT) field [1]. NMT takes a weighted sum of all the annotations to get an expected annotation and focuses only on information relevant to the generation of next target word. A recent work, DeepIntent [26] applies attention in the context of search advertising. Similar to NMT, they use RNN[24] to model text, then learn one global hidden vector to help paying attention on the key words in each query. It is shown that the use of attention can help capturing the main intent of query or ad. DIN designs a local activation unit to soft-search for relevant user behaviors and takes a weighted sum pooling to obtain the adaptive representation of user interests with respect to a given ad. The user representation vector varies over different ads, which is different from DeepIntent in which there is no interaction between ad and user.

We make code publicly available, and further show how to successfully deploy DIN in one of the world's largest advertising systems with novel developed techniques for training large scale deep networks with hundreds of millions of parameters.

3 BACKGROUND

In e-commerce sites, such as Alibaba, advertisements are natural goods. In the rest of this paper, without special declaration, we regard ads as goods. Figure 1 briefly illustrates the running procedure

[1]Experiment code on two public datasets is available on GitHub: https://github.com/zhougr1993/DeepInterestNetwork

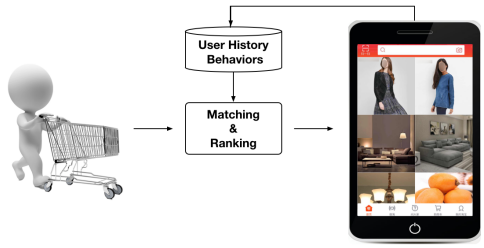

Figure 1: Illustration of running procedure of display advertising system in Alibaba, in which user behavior data plays important roles.

of display advertising system in Alibaba, which consists of two main stages: i) matching stage which generates list of candidate ads relevant to the visiting user via methods like collaborative filtering, ii) ranking stage which predicts CTR for each given ad and then selects top ranked ones. Everyday, hundreds of millions of users visit the e-commerce site, leaving us with lots of user behavior data which contributes critically in building matching and ranking models. It is worth mentioning that users with rich historical behaviors contain diverse interests. For example, a young mother has browsed goods including woolen coat, T-shits, earrings, tote bag, leather handbag and children's coat recently. These behavior data give us hints about her shopping interests. When she visits the e-commerce site, system displays a suitable ad to her, for example a new handbag. Obviously the displayed ad only matches or activates part of interests of this mother. In summary, interests of user with rich behaviors are ***diverse*** and could be ***locally activated*** given certain ads. We show later in this paper making use of these characteristics plays important role for building CTR prediction model.

4 DEEP INTEREST NETWORK

Different from sponsored search, users come into display advertising system without explicitly expressed intentions. Effective approaches are required to extract user interests from rich historical behaviors when building the CTR prediction model. Features that depict users and ads are the basic elements in the CTR modeling of advertisement system. Making use of these features reasonably and mining information from them are critical.

4.1 Feature Representation

Data in industrial CTR prediction tasks is mostly in a multi-group categorial form, for example, [weekday=Friday, gender=Female, visited_cate_ids={Bag,Book}, ad_cate_id=Book], which is normally transformed into high-dimensional sparse binary features via encoding [4, 19, 21]. Mathematically, encoding vector of i-th feature group is formularized as $\mathbf{t}_i \in R^{K_i}$. K_i denotes the dimensionality of feature group i, which means feature group i contains K_i unique ids. $\mathbf{t}_i[j]$ is the j-th element of $\mathbf{t}_i$ and $\mathbf{t}_i[j] \in \{0, 1\}$. $\sum_{j=1}^{K_i} \mathbf{t}_i[j] = k$. Vector $\mathbf{t}_i$ with $k = 1$ refers to one-hot encoding and $k > 1$ refers to multi-hot encoding. Then one instance can be represent as $\boldsymbol{x} = [\boldsymbol{t}_1^T, \boldsymbol{t}_2^T, ...\boldsymbol{t}_M^T]^T$ in a group-wise manner, where M is number of feature groups, $\sum_{i=1}^{M} K_i = K$, K is dimensionality of the entire feature space. In this way, the aforementioned instance with four groups of features are illustrated as:

$$\underbrace{[0, 0, 0, 0, 1, 0, 0]}_{\text{weekday=Friday}} \quad \underbrace{[0, 1]}_{\text{gender=Female}} \quad \underbrace{[0, .., 1, ..., 1, ...0]}_{\text{visited_cate_ids={Bag,Book}}} \quad \underbrace{[0, .., 1, ..., 0]}_{\text{ad_cate_id=Book}}$$

The whole feature set used in our system is described in Table 1. It is composed of four categories, among which user behavior features are typically multi-hot encoding vectors and contain rich information of user interests. Note that in our setting, there are no combination features. We capture the interaction of features with deep neural network.

Table 1: Statistics of feature sets used in the display advertising system in Alibaba. Features are composed of sparse binary vectors in the group-wise manner.

Category	Feature Group	Dimemsionality	Type	#Nonzero Ids per Instance
User Profile Features	gender	2	one-hot	1
	age_level	~ 10	one-hot	1
	...	...	...	...
User Behavior Features	visited_goods_ids	$\sim 10^9$	multi-hot	$\sim 10^3$
	visited_shop_ids	$\sim 10^7$	multi-hot	$\sim 10^3$
	visited_cate_ids	$\sim 10^4$	multi-hot	$\sim 10^2$
Ad Features	goods_id	$\sim 10^7$	one-hot	1
	shop_id	$\sim 10^5$	one-hot	1
	cate_id	$\sim 10^4$	one-hot	1
	...	...	...	...
Context Features	pid	~ 10	one-hot	1
	time	~ 10	one-hot	1
	...	...	...	...

4.2 Base Model(Embedding&MLP)

Most of the popular model structures [3, 4, 21] share a similar Embedding&MLP paradigm, which we refer to as base model, as shown in the left of Fig.2. It consists of several parts:

Embedding layer. As the inputs are high dimensional binary vectors, embedding layer is used to transform them into low dimensional dense representations. For the i-th feature group of $\boldsymbol{t}_i$, let $\mathrm{W}^i = [w_1^i, ..., w_j^i, ..., w_{K_i}^i] \in \mathbb{R}^{D \times K_i}$ represent the i-th embedding dictionary, where $w_j^i \in R^D$ is an embedding vector with dimensionality of D. Embedding operation follows the table lookup mechanism, as illustrated in Fig.2.

- If $\boldsymbol{t}_i$ is one-hot vector with j-th element $\boldsymbol{t}_i[j] = 1$, the embedded representation of $\boldsymbol{t}_i$ is a single embedding vector $\boldsymbol{e}_i = w_j^i$.
- If $\boldsymbol{t}_i$ is multi-hot vector with $\boldsymbol{t}_i[j] = 1$ for $j \in \{i_1, i_2, ..., i_k\}$, the embedded representation of $\boldsymbol{t}_i$ is a list of embedding vectors: $\{\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, ...\boldsymbol{e}_{i_k}\} = \{w_{i_1}^i, w_{i_2}^i, ...w_{i_k}^i\}$.

Pooling layer and Concat layer. Notice that different users have different numbers of behaviors. Thus the number of non-zero values for multi-hot behavioral feature vector $\boldsymbol{t}_i$ varies across instances, causing the lengths of the corresponding list of embedding vectors to be variable. As fully connected networks can only handle fixed-length inputs, it is a common practice [3, 4] to transform the list of embedding vectors via a pooling layer to get a fixed-length vector:

$$\boldsymbol{e}_i = \text{pooling}(\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, ...\boldsymbol{e}_{i_k}). \tag{1}$$